

Slide 2 Reading Notes

Data analysis snapshot | Meds Consulting LLC

Opening

This slide summarizes our initial data analysis. We checked four areas before deeper modeling: missing data, outliers, trend patterns, and early correlations with 30-day readmission.

The top row gives the baseline. We analyzed 12,000 patients, and 43.2 percent were readmitted within 30 days. The data is large enough to support analysis, but it has quality issues that affect interpretation.

1. Missing data

The largest missing data issue is follow-up status. It is missing for 2,587 patients, or 21.6 percent of the dataset. This matters because follow-up status is directly tied to care transitions, which is one of the major areas NHN wants to improve.

Other missing fields are much smaller, usually around 1.5 to 2.0 percent. Those fields can be handled through cleaned or imputed values, but follow-up status needs extra attention because the missingness itself may reveal a workflow visibility gap.

2. Outliers and financial validation

The outlier checks show 191 length-of-stay outliers and 72 total-care-cost outliers. These are important because unusually long stays or unusually high costs can affect the apparent savings opportunity.

We also found 49 records with negative penalty cost. That is a small share of the file, but it should be validated before final financial conclusions because penalty and cost fields feed directly into ROI estimates.

Check	Scale	Meaning
Missing follow-up status	2,587 rows	Care transition visibility gap
Length-of-stay outliers	191 rows	Review extended stays
Total care cost outliers	72 rows	Protect ROI estimates
Negative penalty cost	49 rows	Validate financial extract

3. Trend pattern by high-risk segment

The center chart shows where readmission risk concentrates. The highest observed readmission rates appear in clinically intuitive segments: patients with five or more chronic conditions, patients with unknown follow-up status, patients with three or more prior admissions, skilled nursing discharges, and emergency admissions.

This supports the idea that readmission risk is not evenly distributed across the patient population. The practical implication is that NHN should prioritize targeted care coordination instead of spreading support evenly across all patients.

4. Correlation signals

The correlation chart shows that prior admissions and chronic conditions have the strongest positive relationship with readmission. Length of stay and age also show positive relationships, but they are weaker.

These correlations do not prove causation. They help us decide which variables deserve more attention in regression analysis and segmentation.

Transition cue

After this slide, move from what the data shows to how we will analyze it.